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Regulation: R18**Course Code: 18BP096-4**

IV B.Pharm II Semester Regular Examinations – May, 2025

ADVANCED INSTRUMENTATION TECHNIQUES**Time: 3 hours****(B.PHARM)****Max. Marks: 75M****Answer the following questions****20x1M=20M**

- Which of the following main component of mass spectroscopy deals with resolving the ions into their characteristic mass components according to their mass-to-charge ratio?
 - Ion source
 - Analyzer
 - Detector system
 - Analyzer tube
- Which substance serves as the calibration standard in NMR spectroscopy?
 - DMSO
 - TMS
 - Chloroform
 - Ethanol
- Which of the following technique is used to measure heat flow during a thermal event?
 - TGA
 - DSC
 - UV Spectroscopy
 - IR Spectroscopy
- Which regulatory guidelines are followed for instrument calibration and validation?
 - WHO
 - ICH
 - CPCSEA
 - NABL
- What is the minimum amount of antigen (in micrograms) that can be detected using a Radioimmunoassay?
 - 0.001 µg/ml
 - 0.1 µg/ml
 - 0.01 µg/ml
 - 1 µg/ml
- MALDI is a type of:
 - Chromatographic technique
 - Ionization technique
 - Extraction method
 - Thermal analysis
- In NMR spectroscopy, chemical shift is measured in:
 - ppm
 - Hz
 - Tesla
 - Gauss
- TGA stands for:
 - Total Gas Analysis
 - Thermo-Gravimetric Analysis
 - Thermo-Gain Analysis
 - Thermal Gradient Assessment
- Which principle governs the operation of UV-Visible spectroscopy?
 - Newton's Law
 - Charles' Law
 - Beer-Lambert Law
 - Avogadro's Law
- ICH stands for:
 - International Committee on Health
 - International Conference on Harmonization
 - International Council for Harmonization
 - International Commission for Health
- Which of the following solvents is most commonly used in liquid-liquid extraction?
 - Acetone
 - Water
 - Hexane
 - NaOH
- The solid phase in SPE typically refers to:
 - Solvent phase
 - Stationary phase
 - Mobile phase
 - Volatile phase

13. Who discovered the mass spectrometer?
 - a) Francis Aston
 - b) J. J Thomson
 - c) Ernest O. Lawrence
 - d) Walter Kaufmann
14. GC-MS analysis is suited for analyzing which kinds of compounds?
 - a) Volatile substances
 - b) Non-volatile solids
 - c) Heavy metals
 - d) Crystalline compounds
15. Who introduced the method of Radioimmunoassay?
 - a) Yalow and Berson
 - b) Eva Engvall and Peter Perlman
 - c) Fahey and McKelvey
 - d) None of these
16. What is the main role of the mass spectrometer in HPTLC-MS?
 - a) Separate components
 - b) Detect fluorescence
 - c) Determine mass-to-charge ratio
 - d) Measure boiling point
17. Which of the following is the most commonly used interface?
 - a) Nebulizer
 - b) Chopper
 - c) Filter
 - d) Vaporizing chamber
18. RIA is based on:
 - a) Antigen-receptor binding
 - b) Antigen-antibody binding
 - c) Enzyme-antibody reaction
 - d) Hormone-receptor interaction
19. DSC is used to measure
 - a) Electrical conductivity
 - b) Absorption spectrum
 - c) Heat flow difference
 - d) Refractive index
20. Which of the following is a non-destructive analytical technique?
 - a) TGA
 - b) HPLC
 - c) UV-Visible spectroscopy
 - d) GC

SECTION-B

Answer any two of the following

2x10M=20M

21. Write the principle and instrumentation involved in NMR spectroscopy. Add a note on factors affecting shielding and de-shielding in NMR spectroscopy.
22. Explain the principle, instrumentation and applications of Mass Spectrometry.
23. a. Define and derive the Bragg's equation.
b. Explain the principle and working mechanism of DTA.

SECTION-C

Answer any seven of the following

7X5M=35M

24. Explain chemical shift and factors influencing on Chemical shift.
25. Explain the working of Quadrupole and Ion trap mass analyser.
26. Explain any two X-ray diffraction techniques.
27. Discuss the significance and applications of Radioimmunoassay.
28. Explain the principle and procedure involved in liquid-liquid extraction.
29. Write the principle and working of HPLC/MS.
30. Explain the principle involved in GC-MS/MS and mention its applications.
31. Explain in detail about the method validation process according to ICH and USFDA guidelines.
32. Explain calibration of UV-visible Spectrometer.